

Efficient Recycling Subspace Truncation Method for Periodic Small-Signal Analysis

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Abstract—Periodic small-signal analysis is crucial but time-consuming in RF simulation, since it may deal with many frequency points. While the Krylov subspace recycling method has greatly accelerated the simulation, the increasing memory cost in large-scale RF simulation has become a new bottleneck. A remedy for memory shortage is to restart the recycling algorithm, but may cause excessive extra iterations. To address this issue, this paper outlines a framework of recycling subspace truncation method for periodic small-signal analysis, provided with an efficient initial guess choice method and a Floquet-based subspace truncation strategy. Numerical results show that compared to the existing methods, the proposed method achieves up to $2.5\times$ speedup in the same memory cost.

Index Terms—RF simulation, periodic small-signal analysis, Krylov subspace recycling, subspace truncation.

I. INTRODUCTION

PERIODIC small-signal analysis, including PAC (periodic AC), PXF (periodic transfer function), and PNoise (periodic noise), plays a crucial role in understanding the behavior of radio-frequency (RF) circuits operating under repetitive conditions, such as oscillators, switching converters, and communication systems [1]. Periodic small-signal analysis computes the steady-state response of a linear periodically time-varying circuit to a sinusoidal stimulus at different frequencies.

The primary challenge arising in periodic small-signal analysis is how to efficiently solve a sequence of linear systems corresponding to different frequency points [2], [3], because the computational cost of solving all the linear systems increases linearly with the number of frequency points. The authors in [2] transform the original linear systems into a sequence of shifted linear systems, and then the solution of these shifted linear systems can be accelerated using the Krylov subspace recycling method. The recycling generalized conjugate residual (R-GCR) method introduced in [2] is up to forty times faster than the original direct factorization method. A more efficient and robust version of recycling algorithm based on generalized minimal residual (GMRES) method [4] is developed in [3].

However, with the explosive growth of RF circuit scale, the recycled subspace expands fast in frequency sweep, causing excessive amount of memory and computational costs. The issue of memory shortage may diminish the stability and robustness of the previous recycling algorithms. A remedy for

memory shortage is to restart the whole recycling algorithm, but may cause excessive extra iterations. In the field of numerical algebra, the GCRO-DR method [5] is a well-known recycling method with subspace truncation to solve a sequence of linear systems, but this method does not take full advantage of the special properties in periodic small-signal analysis.

In this paper, we outline a framework of recycling subspace truncation method for periodic small-signal analysis to address the issue of memory shortage. Previous subspace recycling methods for periodic small-signal analysis are applicable in our framework. Two core problems in our framework are: (1) how to choose a good initial guess from the last system; (2) what vectors should be retained in the recycled subspace. Inappropriate choice of initial guess and recycled subspace may cause excessive number of additional iterations, or even stagnate the convergence of Krylov subspace iterative methods.

For the first problem, we propose an optimal initial guess recycling method to obtain a better initial guess for the next system. For the second problem, we develop a Floquet-based subspace truncation strategy to reduce the extra iterations caused by subspace truncation. We have observed that the coefficient matrix can be identified as the discretization of the state-transition matrix of a linear periodically time-varying system. The Floquet theory in [6] can be applied to characterize the eigenvalue distribution of the state-transition matrix. Based on the eigenvalue distribution, we reveal that the subspace corresponding to the largest Floquet multipliers significantly affect the convergence of iterative methods, and then we can establish the formulation of our Floquet-based subspace truncation strategy.

The paper is organized as follows. In Section II, we review the derivation of the shifted linear systems in periodic small-signal analysis, and then present our observation of the eigenvalue distribution of the coefficient matrix based on the Floquet theory. In Section III, we outline a framework of recycling subspace truncation method for periodic small-signal analysis, provided with our optimal initial guess recycling method and Floquet-based subspace truncation strategy. In Section IV, we apply the proposed method to several examples to demonstrate the effectiveness of our method. Finally, conclusions are given in Section V.

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II. SHIFTED LINEAR SYSTEMS IN SMALL-SIGNAL ANALYSIS

A. Derivation

Using modified nodal analysis [7], the RF circuit can be characterized by the differential equations

$$\frac{d}{dt}(q(v(t))) + i(v(t)) + u(t) = 0, \quad (1)$$

where all the vectors are in $\mathbb{R}^N$, $v(t)$ is the node voltages, and $i(v(t))$, $q(v(t))$ are the resistive node currents and node charges respectively.

Suppose the periodic operating point $v_L(t)$ (with period T_L) has been computed by the periodic steady-state (PSS) analysis [1]. Linearizing the circuit equations (1) around $v_L(t)$ yields the linear periodically time-varying system of small-signal response $v_s(t)$ [2],

$$\frac{d}{dt}(C(v_L(t))v_s(t)) + G(v_L(t))v_s(t) + u_s(t) = 0, \quad (2)$$

where

$$C(v_L(t)) = \left. \frac{dq}{dv} \right|_{v=v_L(t)}, \quad G(v_L(t)) = \left. \frac{di}{dv} \right|_{v=v_L(t)}.$$

From the theory of periodically time-varying systems [6], it is known that the steady-state small-signal response $v_s(t)$ satisfies the periodic boundary condition

$$v_s(t + T_L) = v_s(t)\sigma(f_s), \quad \sigma(f_s) = \exp(2\pi j(f_s/f_L)).$$

Now we consider solving $v_s(t)$ using time-domain discretization methods (backward Euler, Gear2 or others). Given a partition of the interval $[0, T_L]$ as $0 = t_0 < t_1 < \dots < t_M = T_L$. Let $h_i = t_i - t_{i-1}$, $C_i = C(v_L(t_i))$ and $G_i = G(v_L(t_i))$. For each t_i , the approximate value of $v_s(t_i)$ is denoted by $v_{s,i}$. For simplicity, we use backward Euler method to discretize (2) at t_i ($1 \leq i \leq M$). After assembling the M equations at each time point, we can obtain the linear systems

$$(\sigma(f_s)L + B) \begin{bmatrix} v_{s,1} \\ v_{s,2} \\ \vdots \\ v_{s,M} \end{bmatrix} = -\sigma(f_s)u_s(f_s), \quad (3)$$

where

$$L = \begin{bmatrix} \frac{C_1}{h_1} + G_1 & & & \\ -\frac{C_1}{h_2} & \frac{C_2}{h_2} + G_2 & & \\ & \ddots & \ddots & \\ & & -\frac{C_{M-1}}{h_M} & \frac{C_M}{h_M} + G_M \end{bmatrix},$$

$$B = \begin{bmatrix} 0 & \dots & 0 & -\frac{C_M}{h_1} \\ 0 & \dots & 0 & 0 \\ \vdots & & \vdots & \vdots \\ 0 & \dots & 0 & 0 \end{bmatrix}.$$

Take L as the preconditioner and transform the original linear system to a shifted linear system

$$(\sigma(f_s)I_{MN} + L^{-1}B)v_s = -\sigma(f_s)L^{-1}u_s(f_s).$$

Note that the coefficient matrix $\sigma I_{MN} + L^{-1}B$ has the block structure

$$\sigma I + L^{-1}B = \begin{bmatrix} \sigma I_N & \dots & 0 & -P_1 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \dots & \sigma I_N & -P_{M-1} \\ 0 & \dots & 0 & \sigma I_N - P_M \end{bmatrix}. \quad (4)$$

Hence, one can first solve $v_{s,M}$ from the shifted linear system

$$(\sigma(f_s)I_N - P_M)v_{s,M} = -\sigma(f_s)(L^{-1}u_s(f_s))_M. \quad (5)$$

Then the remaining $v_{s,i}$ for $i = M-1, M-2, \dots, 1$ can be obtained through backward substitution. After simplifying the notations, the main task of the periodic small-signal solver is to efficiently solve a sequence of shifted linear systems

$$(\sigma^{(k)}I_N - P_M)x = b^{(k)}, \quad \sigma^{(k)} = \exp(2\pi j(f_s^{(k)}/f_L)),$$

corresponding to a sequence of frequency points $f_s^{(k)}$.

B. Relation between P_M and state-transition matrix

Denote $\Phi(t, s)$ as the state-transition matrix of the linear periodically time-varying system of DAEs

$$\frac{d}{dt}(C(t)v(t)) + G(t)v(t) = 0. \quad (6)$$

Note that (6) is exactly the homogeneous part of (2).

Suppose we are now required to perform a transient simulation on the homogeneous system (6) with the initial condition $\tilde{v}(0)$. Using the same discretization method as in Section II-A, the approximate solution $\tilde{v}(t)$ at $t = t_i$ can be obtained as

$$\tilde{v}(t_i) = P_i \tilde{v}(0), \quad 1 \leq i \leq M,$$

where P_i is defined in (4). In fact, performing one preconditioner L on (3) can be thought of as performing a transient simulation over one period.

In particular, we have $\tilde{v}(T_L) = P_M \tilde{v}(0)$. Hence, P_M can be considered as the discretization of the state-transition matrix $\Phi(T_L, 0)$. Therefore, it can be predicted that if the time step h_i is small enough, P_M may have some properties similar to the state-transition matrix $\Phi(T_L, 0)$.

C. Eigenvalue distribution of P_M

From the Floquet theory of linear periodically time-varying systems of DAEs [6], under some assumptions the state-transition matrix $\Phi(t, s)$ for (6) can be written as

$$\Phi(t, s) = U(t)D(t-s)V(s)^*C(s)$$

where $U(t), V(s) \in \mathbb{C}^{N \times r}$ are T_L -periodic matrices of rank r , and $D(t) = \text{diag}\{\exp(\rho_1 t), \dots, \exp(\rho_r t)\}$ with all $\rho_i \neq 0$. Here $\rho_1, \dots, \rho_r$ are called Floquet exponents, and $\lambda_i = \exp(\rho_i T_L)$ are defined as Floquet multipliers. The r Floquet multipliers are exactly all the nonzero eigenvalues of the state-transition matrix $\Phi(T_L, 0)$.

It can be shown (e.g. [8]) that if one of the Floquet multipliers is 1, and the magnitude of other $r-1$ Floquet multipliers is less than 1, then (6) has a asymptotically orbitally stable solution $\dot{v}_L(t)$. All practical stable oscillators satisfy this

condition [6], i.e. all the nonzero eigenvalues of $\Phi(T_L, 0)$ have magnitude less than 1 except one eigenvalue equal to 1.

Since the coefficient matrix P_M can be identified as the discretization of $\Phi(T_L, 0)$, the eigenvalues of P_M may well approximate the Floquet multipliers. Hence, we can predict that P_M may have similar eigenvalue distribution to $\Phi(T_L, 0)$. In numerical experiments, we observe that the maximal magnitude of eigenvalues of P_M is close to 1 (often less than 1) and P_M may have several eigenvalues near 1 (see Section IV-A).

III. RECYCLING SUBSPACE TRUNCATION METHOD FOR PERIODIC SMALL-SIGNAL ANALYSIS

A. Previous Krylov Subspace Recycling Methods

We briefly review the previous Krylov subspace recycling methods for periodic small-signal analysis. The details of the algorithms can be seen in [2], [3].

The Krylov subspace recycling methods for a sequence of shifted linear systems are based on the relation

$$(\sigma_2 I - P) v = (\sigma_1 I - P) v + (\sigma_2 - \sigma_1) v. \quad (7)$$

The relation (7) means that if $(\sigma_1 I - P) v$ has been computed, then $(\sigma_2 I - P) v$ can be obtained just using operations on vectors. Hence, the subspace computed in the last system can be recycled to the next system.

Given recycled subspace U_m , C_m of dimension m satisfying $(\sigma^{(k)} I - P) U_m = C_m$. The R-GCR method proposed in [2] computes the j -dim augmented subspace V_j , $\tilde{V}_j$ satisfying the relation

$$(\sigma^{(k)} I - P) [U_m, V_j] = [C_m, \tilde{V}_j] \quad (8)$$

where we call $[U_m, V_j]$ ($[C_m, \tilde{V}_j]$) the total subspace. The dimension of the total subspace is $m + j$. With the relation (8), the approximate solution $x^{(k)}$ and residual $r^{(k)}$ can be updated as

$$\begin{aligned} y_{m+j} &= [C_m, \tilde{V}_j]^* r_0^{(k)}, \\ x^{(k)} &= x_0^{(k)} + [U_m, V_j] y_{m+j}, \\ r^{(k)} &= r_0^{(k)} - [C_m, \tilde{V}_j] y_{m+j}. \end{aligned}$$

Recently, a more robust and efficient recycling method for periodic small-signal analysis called R-GMRES is developed in [3]. R-GMRES computes the following relation

$$(\sigma^{(k)} I - P) [U_m, V_j] = [C_m, V_{j+1}] \begin{bmatrix} T_m & B_j \\ 0_{(j+1) \times m} & \underline{H}_j \end{bmatrix}. \quad (9)$$

Similar to R-GCR, the approximate solution $x^{(k)}$ and residual $r^{(k)}$ can be updated with the relation (9).

B. A framework of recycling subspace truncation method for periodic small-signal analysis

In the process of R-GCR or R-GMRES, if the total subspace is too large, the issue of memory shortage is encountered. To address the issue, we retain a part of vectors in the recycled subspace and discard others. Here we outline a framework of recycling subspace truncation method for periodic small-signal

analysis. The framework is summarized in Algorithm 1. In our framework, either R-GCR or R-GMRES can be used.

In the whole process of frequency sweep, we set the maximal dimension of recycled subspace and augmented subspace to be $m_{\max}$ and $j_{\max}$.

Before starting to perform the recycling algorithm, if we have computed the total subspace $[U_m, V_j]$ from the last system, then the total subspace should be truncated to a new recycled subspace. Here we still denote the new recycled subspace as U_m . The dimension of recycled subspace U_m after truncation should be no larger than $m_{\max}$.

Before starting to solve each linear system, we need to select a good initial guess $x_0^{(k)}$ from the last system (for the first system, $x_0^{(1)}$ is set to be zero). If the initial guess is chosen appropriately, then the initial residual $r_0^{(k)}$ can be computed from the residual $r^{(k-1)}$ of the last system without any matrix-vector multiplication (see Section III-C).

In the process of solving each shifted linear system, the augmented subspace V_j is computed by R-GMRES or R-GCR, and the total subspace is expanded to $[U_m, V_j]$. If the dimension $m + j$ of the total subspace $[U_m, V_j]$ is larger than $m_{\max} + j_{\max}$, then we also need to truncate the total subspace to a recycled subspace of dimension $m_{\max}$. Then R-GMRES or R-GCR is restarted with the new recycled subspace after truncation.

Algorithm 1 A framework of recycling subspace truncation method for periodic small-signal analysis

Input: U_m, V_j : the recycled subspace and augmented subspace from the last system;

$m_{\max}$: the maximal dimension of recycled subspace;

$j_{\max}$: the maximal dimension of augmented subspace;

tol: tolerance of relative residual.

Output: Approximate solutions $x^{(k)}$ for shifted linear system $(\sigma^{(k)} I - P) x = b^{(k)}$.

- 1: Truncate the total subspace $[U_m, V_j]$ to a new recycled subspace U_m of dimension $m \leq m_{\max}$.
 - 2: Obtain a good initial guess $x_0^{(k)}$ and compute the initial residual $r_0^{(k)}$.
 - 3: **while** $\|r^{(k)}\|_2 / \|b^{(k)}\|_2 > \text{tol}$ **do**
 - 4: Perform the recycling algorithm R-GMRES or R-GCR to compute the augmented subspace V_j , and update the approximate solution $x^{(k)}$ and residual $r^{(k)}$.
 - 5: **if** $m + j > m_{\max} + j_{\max}$ **then**
 - 6: Truncate the total subspace $[U_m, V_j]$ to a new recycled subspace U_m of dimension $m = m_{\max}$.
 - 7: Restart R-GMRES or R-GCR with the truncated recycled subspace.
-

In the entire simulation process, the main memory consumption in storing the total subspace is to store at most $(2m_{\max} + j_{\max} + 1)$ vectors of dimension N for R-GMRES, or $2(m_{\max} + j_{\max})$ vectors for R-GCR.

Two core problems in our framework are: (1) how to choose a good initial guess from the last system; (2) what vectors

should be retained in the new recycled subspace. For the first problem, our optimal initial guess recycling method is discussed in Section III-C. For the second problem, we propose a Floquet-based subspace truncation strategy in Section III-D.

The effectiveness of the recycling method may depend on the order in which the small-signal frequencies are arranged. In our numerical experiments (see Section IV), for simplicity, the small signal frequencies f_s are arranged in ascending order. An effective sweeping order can be seen in [3].

C. Optimal initial guess recycling method

In this section, we discuss how to select a good initial guess. We present our optimal initial guess recycling method.

Suppose we have computed the approximate solution $x^{(k-1)}$ and the residual $r^{(k-1)}$ of the last system. The traditional approach (for example, in GCRO-DR [5]) is to directly take the initial guess $x_0^{(k)} = x^{(k-1)}$. However, in most cases, this is not a good choice. Our idea is to introduce a new parameter α , and take $x_0^{(k)} = \alpha x^{(k-1)}$ as the initial guess. The optimal parameter α is determined s.t. the initial residual $r_0^{(k)}$ is minimized, i.e.

$$\alpha = \arg \min_{\alpha \in \mathbb{C}} \left\| b^{(k)} - \left(\sigma^{(k)} I - P \right) \left(\alpha x^{(k-1)} \right) \right\|_2.$$

Denote $w = \left(\sigma^{(k)} I - P \right) x^{(k-1)}$. From the relation (7), w can be computed only using vector additions and scalar multiplications as

$$w = \left(b^{(k-1)} - r^{(k-1)} \right) + \left(\sigma^{(k)} - \sigma^{(k-1)} \right) x^{(k-1)}. \quad (10)$$

Then the optimal parameter α can be determined by the least-squares method as

$$\alpha = w^* b^{(k)} / w^* w. \quad (11)$$

The algorithm is described in Algorithm 2.

Algorithm 2 Optimal initial guess recycling method

Input: $x^{(k-1)}$: approximate solution of the last system;
 $r^{(k-1)}$: residual of the last system.

Output: Initial guess $x_0^{(k)}$ and initial residual $r_0^{(k)}$.

- 1: Compute w as in (10).
 - 2: Compute α as in (11).
 - 3: Compute initial guess $x_0^{(k)} = \alpha x^{(k-1)}$
 - 4: Compute initial residual $r_0^{(k)} = b^{(k)} - \alpha w$.
-

Note that the residual $r_0^{(k)}$ is computed without any matrix-vector multiplication and the computational cost of our initial guess method involves only 2 inner-products plus no additional matrix-vector multiplication, but it can save several iterations in iterative algorithms (see Section IV-B).

In fact, since the order of the small-signal frequencies can be arbitrary, the initial guess gained from the last system may not be the best. A better choice is to obtain the initial guess from the computed system corresponding to the nearest small-signal frequency.

D. Floquet-based subspace truncation strategy

In this section, we discuss our Floquet-based subspace truncation strategy. We have observed that inappropriate subspace truncation strategy may result in excessive additional iterations. Hence, an effective truncation strategy is required.

The convergence of Krylov subspace methods depends to a large degree on the eigenvalue distribution [9]. Indeed, small eigenvalues may greatly stagnate the convergence. Hence, in GCRO-DR [5], the authors retain the harmonic Ritz vectors corresponding to the smallest harmonic Ritz values. This strategy is applicable to general cases, but for the shifted linear systems arising in small-signal analysis, a better truncation strategy can be obtained based on Floquet multipliers.

From the observation in Section II-C, P_M has some eigenvalues λ whose magnitude $|\lambda| \approx 1$. Since $\sigma(f_s) = \exp(2\pi j(f_s/f_L))$, then for some small-signal frequency f_s , the coefficient matrix $\sigma(f_s)I - P$ may have eigenvalues near 0. In this situation, $\sigma(f_s)I - P$ is ill-conditioned, and the Krylov subspace iterative methods may need more iterations to converge.

Returning to the original RF circuits, these eigenvalues close to 1 correspond to slow time constants [10]. In the circuit with slow time constants, the response may be mixed with slowly decaying parts. Hence, the iterative methods need more iterations to detect these parts.

From the above analysis, our idea is to retain in the recycled subspace the eigenvectors corresponding to the eigenvalues with magnitude near 1. In Section II-C, we have observed that the maximal magnitude of the eigenvalues of P_M is near 1. Hence, these eigenvalues with magnitude near 1 are exactly the largest eigenvalues of P_M (which correspond to the largest Floquet multipliers). The largest eigenvalues can be computed by the Ritz values [11] during the frequency sweep. The detailed algorithm is presented as follows.

Here we take R-GMRES as an example. Suppose we have obtained the relation (9), then the QZ algorithm [12] (for the case where $m = 0$, the QR algorithm [13], [14] is enough) is applied to compute all the eigenpairs (z_i, μ_i) of the matrix pencil (E, F) where

$$E = \begin{bmatrix} (U_m^* C_m) T_m & (U_m^* C_m) B_m + (U_m^* V_{j+1}) \underline{H}_j \\ 0_{j \times m} & H_j \end{bmatrix}, \quad (12)$$

$$F = \begin{bmatrix} I_m & U_m^* V_j \\ (U_m^* V_j)^* & I_j \end{bmatrix}.$$

Then choose the $\tilde{m}$ eigenvectors corresponding to the eigenvalues with largest $|\sigma^{(k)} - \mu_i|$. The $\tilde{m}$ eigenvectors are stored in $Z_{\tilde{m}} \in \mathbb{C}^{(m+j) \times \tilde{m}}$. The new recycled subspace $U_{\tilde{m}}$ can be computed with $Z_{\tilde{m}}$. The algorithm of our Floquet-based subspace truncation strategy is described in Algorithm 3.

The main computational cost of our Floquet-based strategy lies in computing $U_m^* C_m$, $U_m^* V_{j+1}$ and a $(m+j)$ -dim generalized eigenvalue problem. Since m and j are usually much smaller than N , the computational cost is a very small portion of the whole solving process.

Algorithm 3 Floquet-based subspace truncation strategy

Input: $\tilde{m}$: dimension of recycled subspace after truncation;
 U_m, C_m : the recycled subspace;
 V_{j+1} : the augmented subspace.

Output: Recycled subspace after truncation: $U_{\tilde{m}}, C_{\tilde{m}}$ satisfying $(\sigma^{(k)}I - P)U_{\tilde{m}} = C_{\tilde{m}}$.

- 1: Compute all the eigenpairs (z_i, μ_i) of the matrix pencil defined in (12).
- 2: Choose the $\tilde{m}$ eigenvectors corresponding to the eigenvalues with the largest $|\sigma^{(k)} - \mu_i|$ and store them in $Z_{\tilde{m}}$.
- 3: Compute $U_{\tilde{m}} = [U_m, V_j]Z_{\tilde{m}}$.
- 4: Compute $C_{\tilde{m}} = [U_m, V_{j+1}] \begin{pmatrix} T_m & B_j \\ 0 & H_j \end{pmatrix} Z_{\tilde{m}}$.

IV. NUMERICAL EXPERIMENTS

We have tested our proposed method using MATLAB. Test cases are provided by our industrial partner, as shown in Table I. The “Discretization” column in the table describes the discretization scheme. BE(1) means that the backward Euler scheme is applied only to the first time point, while BE(1, 2) applies to the first and second time point. For the accuracy of simulation, Gear2 scheme is applied to other time points. In AC analysis, the frequency sweep usually comes in three types: linear (LIN), octave (OCT) and decade (DEC) [15]. We focus on LIN and DEC in our numerical experiments.

TABLE I
DETAILS OF THE TEST CASES

Index	# Timesteps	# Nodes	Discretization
case 1	2003	3886	BE(1,2), Gear2
case 2	2451	9482	BE(1,2), Gear2
case 3	729	16224	BE(1,2), Gear2
case 4	801	3058	BE(1), Gear2

In practical computation, the matrix-vector multiplication $P * x$ is implicitly performed. The LU factorization of the diagonal blocks of L is precomputed, so the runtime we present does not include this portion.

All the numerical experiments are implemented in MATLAB R2020b with Intel Xeon Gold 6226R @2.90 GHz. For all the linear systems, the tolerance of relative residual is set to be 10^{-8} .

A. Eigenvalue distribution of P_M

To verify our prediction of the eigenvalue distribution of P_M in Section II-C, we compute all the eigenvalues of P_M using `eigs` function in MATLAB. The results are shown in Table II. It is observed that the maximal magnitude of P_M ’s eigenvalues are close to 1 in four test cases. Additionally, P_M has several eigenvalues close to 1, causing that $\sigma(f_s)I - P_M$ has several eigenvalues close to 0 when f_s is near an integer multiple of f_L . Since almost all the eigenvalues of P_M have magnitude less than 1, the eigenvalues near 1 are the ones with the largest magnitude. The results of the numerical experiments are in agreement with our previous prediction in Section II-C.

TABLE II
THE EIGENVALUE DISTRIBUTION OF P_M

Index	#Nodes	$ \lambda _{\max}$	# $ \lambda - 1 \leq 0.05$
case 1	3886	1.0000	4
case 2	9482	0.9864	1
case 3	16224	1.0000	18
case 4	3058	1.0000	48

B. Choice of initial guess

We demonstrate the effectiveness of our optimal initial guess recycling method. We test three different initial guess choices by performing GMRES method with no recycled subspace on case 4. The sweeping type is chosen as “LIN 201 0 f_L ”. The results are shown in Figure 1. “Zero” means setting the initial guess to be zero, while “Original” means taking the approximate solution of the previous system as the initial guess for the next system. Our method is referred to as “Proposed”.

From Figure 1, the number of iterations required for “Original” and “Zero” is almost the same, though at some frequency points “Original” may require more steps to converge. Hence, it is shown that taking the solution of the previous system as the initial guess is not a good choice. However, with our proposed method, convergence iterations can be greatly reduced over a wide range of small-signal frequencies. When f_s is far from an integer multiple of f_L , the convergence of iterative methods is really good, and our proposed method can still save several iterations.

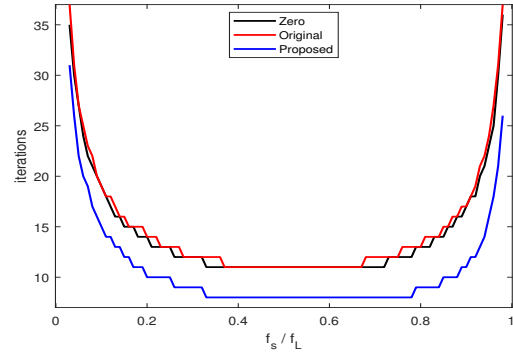


Fig. 1. Number of iterations for case 4 with different initial guess choices

C. Subspace truncation strategy

This section demonstrates the effectiveness of our Floquet-based subspace truncation strategy. We compare our Floquet-based truncation strategy with the deflated restarted (DR) method used in GCRO-DR [5], [9]. We test two subspace truncation methods in our framework of algorithm on case 4 for the sweeping type “DEC 20 10 10^7 ”. The maximal dimension of recycled subspace is set to be 25. To better highlight the difference between two methods, the initial guess is set to be zero. The results are shown in Figure 2. “DR” means using deflated restarted method while “Floquet” is our Floquet-based subspace truncation method.

TABLE III
COMPARISON BETWEEN GCRO-DR AND OUR PROPOSED METHOD FOR “DEC 6 10 10⁸” (TIME UNIT: SECOND).

Index	$m_{\max} = 20, j_{\max} = 40$					$m_{\max} = 25, j_{\max} = 30$				
	GCRO-DR		Proposed			GCRO-DR		Proposed		
	#mvm	Time	#mvm	Time	Speedup	#mvm	Time	#mvm	Time	Speedup
case 1	110	293.50	79	180.98	1.62	74	202.37	57	133.70	1.51
case 2	189	2029.30	112	1026.15	1.98	75	1073.05	51	463.52	2.32
case 3	223	1269.17	103	565.55	2.24	175	1027.06	79	439.68	2.34
case 4	2239	1579.53	1700	1222.87	1.29	2095	1480.24	1537	1099.54	1.35

TABLE IV
COMPARISON BETWEEN GCRO-DR AND OUR PROPOSED METHOD FOR “LIN 41 0 f_L ” (TIME UNIT: SECOND).

Index	$m_{\max} = 20, j_{\max} = 40$					$m_{\max} = 25, j_{\max} = 30$				
	GCRO-DR		Proposed			GCRO-DR		Proposed		
	#mvm	Time	#mvm	Time	Speedup	#mvm	Time	#mvm	Time	Speedup
case 1	281	690.91	145	321.70	2.15	245	610.04	106	240.86	2.53
case 2	277	2874.33	184	1729.40	1.66	244	2536.48	105	957.53	2.65
case 3	322	1796.86	197	1058.18	1.70	279	1580.90	153	817.34	1.93
case 4	1220	845.26	811	562.13	1.50	1108	770.82	792	553.93	1.39

From Figure 2, a noticeable reduction in iterations is observed for our proposed method. When f_s is very small (near 10 Hz), the DR method requires significantly more iterations than our method. During the process of frequency sweep, DR may experience sudden increases of iterations, while this phenomenon does not occur in our method.

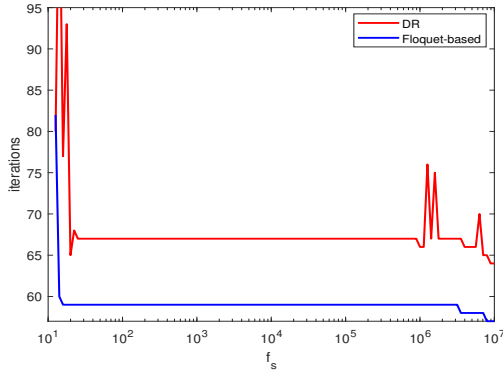


Fig. 2. Number of iterations for case 4 with different subspace truncation methods

D. Comparing proposed method with GCRO-DR

We compare our recycling subspace truncation method for periodic small-signal analysis with GCRO-DR [5]. Two types of frequency sweeps are tested: “DEC 6 10 10⁸” and “LIN 41 0 f_L ”. For both frequency sweeps, the frequency points are arranged in ascending order. To better reflect the real simulation process, we choose $m_{\max}$ and $j_{\max}$ such that the memory consumption for storing the total subspace remains the same. The numerical results are shown in Table III and Table IV. “#mvm” means the total number of matrix-vector

multiplications, which is exactly the number of iterations in the whole algorithm. “Speedup” is the ratio of the runtime.

From Table III and Table IV, it can be observed that for both GCRO-DR and our proposed method, $m_{\max} = 25$ is faster than $m_{\max} = 20$, which means appropriately increasing the maximum dimension of recycled subspace can reduce the total number of matrix-vector multiplications. Compared to GCRO-DR, our proposed method can significantly reduce the simulation time of both frequency sweeps. Our proposed method can achieve up to a 2.65 $\times$ speedup, and when the maximal dimension of recycled subspace increases, the speedup of our proposed method is better. Considering the magnitudes of speedup, the performance of our proposed method is very competent.

V. CONCLUSION

In this paper, we present a framework of recycling subspace truncation method for periodic small-signal analysis. For initial guess, we propose an efficient optimal initial guess recycling method, which can save many iterations with low computational cost. For subspace truncation, we observe that the subspace corresponding to the largest Floquet multipliers significantly affect the convergence of iterative methods. Based on the observation, we develop the Floquet-based subspace truncation strategy. The effectiveness of our proposed method is verified by the numerical experiments. We hope that our proposed method can help overcome the difficulty of memory shortage encountered in periodic small-signal analysis for large-scale circuit designs. Possible future improvement would be to find a better sweeping order of small-signal frequencies.

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